Riko Miyagi

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EDUCATION

University of Waterloo: Bachelor of Environmental Studies, Geospatial Data Science *(Co-op)*

TECHNICAL SKILLS

Software/Tools: Excel, SQL, Python, R, Tableau, Power BI, BigQuery, SQLite, ArcGIS Pro, QGIS, SNAP, HTML, DrRacket, Microsoft & Google Ecosystem, GitHub, Jupyter

EXPERIENCE

Energy Coordinator - Ecobee, Toronto On Sep - Dec 2026

- Designed and developed an automated Power BI dashboard to streamline thermostat footprint count requests, reducing BigQuery query execution enabling faster response times across utility demand response programs.
- Conducted device analyses using BigQuery SQL to support utility partner program eligibility
- Prepared regular reports and dashboards on sales performance, CRM metrics, KPI achievement, providing actionable insights and recommendations to drive performance improvement

Energy Reporting and Measurement/Verification Assistant - Parity Inc, Toronto On Jan-Apr 2026

- Measured and verified energy and emissions savings from AI-driven HVAC optimization across 200+ commercial buildings in North America by collecting and analyzing building data.
- Produced energy savings reports including baseline comparisons and performance analyses to support guaranteed savings commitments for client portfolios.
- Detected and resolved anomalies in utility billing data across client portfolios; built automation tools in using AppScripts to streamline auditing workflows, reducing manual review time.

PROJECTS

Heat Pump vs. Gas Furnace Feasibility Study Feb 2026

- Developed a Python-based financial and environmental analysis tool comparing the 15-year total cost of ownership between air-source heat pumps and natural gas furnaces for Ontario homes.
- Modeled energy consumption using Ontario energy rates, and government rebates, finding a $9,500 cost difference and 94.5% emissions reduction in favor of heat pumps environmentally.

Smart Thermostat Savings Calculator Oct 2026

- Built an interactive calculator modeling smart thermostat savings across 40 cities using the degree-day method, adjusted for home size, insulation, and heating system.
- Applied DOE temperature-setback methodology to estimate savings in dollars and emissions avoided, using province-specific utility rates and grid carbon intensity.
- Added a payback-period calculator and rate override, validating logic against test cases before deployment.

Grid Carbon Intensity Comparison Jun 2026

- Designed and built a relational SQLite database benchmarking electricity grid carbon intensity across five countries using 25 years of Ember's global electricity dataset.
- Wrote SQL queries using joins, conditional aggregation, and window functions (LAG, RANK) to quantify year-over-year change and rank countries by decarbonization performance.

Japan's Post-Fukushima Energy Transition (2000-2024) Sept 2026

- Built an interactive Tableau dashboard analyzing Japan's electricity generation using U.S. EIA international energy data, visualizing the shift in fuel sources following the 2011 Fukushima accident.
- Found that nuclear's average share of generation fell from 26% pre-Fukushima to 3.7% post-Fukushima, while non-hydro renewable generation grew 4.6x over the same period.

CERTIFICATIONS

University of Michigan Python Coursera Course (Core Python programming) Aug 2025
Google Data Analytics Professional Certificate (Spreadsheets, SQL, R, Tableau) Aug 2025
IBM Data Science Certificate (SQL, Python, R, GitHub, Jupyter) Oct 2026